Causal Check on Othello-GPT: Separating Downstream World-Model Computation from Direct Readout Bias

Rohan Gupta · Purdue University (gupt1156@purdue.edu)

Physical Reasoning Research Take-Home · Direction 3 (What Models Represent) — Prompt 4: Causal Check

Abstract

Linear probes and activation steering are standard tools for certifying that sequence models have learned internal world models, yet both face two fundamental causal confounds: (1) *skip-connection readout bias*, where injecting a steering vector $\boldsymbol{\delta}_\ell$ into intermediate residual streams also adds $\sum_\ell \boldsymbol{\delta}_\ell$ directly to the final unembedding W_U, and (2) *downstream self-repair*, where high single-layer probe accuracy fails to translate into single-layer causal control. Auditing the 8-layer `Othello-GPT` model and its 1,001-board counterfactual intervention benchmark (Li et al., 2023; Nanda et al., 2023), we introduce a skip-connection decomposition that subtracts the accumulated steering vector before $\mathrm{LN}_{\text{final}} + W_U$ (`Mediated-Only`) vs. applying it exclusively at the final readout (`Direct-Logit Bypass`). We show that 97.9% of counterfactual error reduction ($2.723 \rightarrow 0.158$) survives zeroing out the direct skip connection, whereas direct readout bias yields zero probability gain ($\Delta P_{\text{new}} = +0.000$) on newly legal moves. Furthermore, auditing the official `mech_int_otheloGPT` codebase reveals three implementation discrepancies (hardcoded Layer-5 probes, pre-MLP `attn_out` hooks, and non-contrastive class addition); aligning the hook site to `resid_post` and steering along the contrastive flip direction $\hat{\mathbf{d}}_s^\ell \propto \mathbf{w}_{s,\text{tgt}}^\ell - \mathbf{w}_{s,\text{src}}^\ell$ reduces counterfactual error by 8.5× ($0.102 \rightarrow \mathbf{0.012}$, or 12 errors across 1,001 boards).

1 The Question and Why It Matters Now

A central claim in world-model interpretability is that self-supervised sequence models learn linear internal representations of world state that causally drive predictions. In `Othello-GPT`, Nanda et al. (2023) showed that probing residual stream activations in player-relative coordinates (EMPTY, YOURS, MINE) yields $> 99\%$ linear accuracy by Layer 4, and that adding probe vectors across layers steers the model into predicting legal moves for a counterfactual board B'. However, two fundamental questions remain unresolved whenever linear steering alters a transformer's output:

1. **Mediated world-model computation vs. direct readout bias:** Because a transformer's residual stream forms an identity skip connection to the final unembedding ($\mathbf{x}_L = \mathbf{x}_\ell + \sum_{k=\ell}^{L-1} f_k(\mathbf{x}_k)$), injecting a steering vector $\boldsymbol{\delta}_\ell$ across layers $\ell \in \{2, \ldots, 7\}$ also adds $\sum_\ell \boldsymbol{\delta}_\ell$ directly into the final readout W_U (Figure 1a). Does steering succeed because intermediate Attention and MLP layers recompute multi-ray Othello flanking rules over the edited board state, or merely because $\boldsymbol{\delta}$ acts as a direct linear logit bias at W_U?
2. **Single-layer decodability vs. causal control:** If a linear probe decodes the board state with 99.4% accuracy at Layer 5, does intervening at Layer 5 alone flip the model's move predictions, or do downstream layers overwrite single-layer state edits by attending back to unmodified history tokens (Figure 1b)?

Why it matters for world models now. Linear probe R^2 and activation steering are widely used to certify that video, robotics, and physics models have learned actionable world representations. If a representation is 99% decodable at layer ℓ yet causally overwritten unless edited across a specific circuit span—or if steering metrics are confounded by direct skip-connection readout bias—standard probing diagnostics will systematically overstate whether a learned world model can support counterfactual planning.

2 Method: Models, Data, Metrics, and Controls

Model and Data. We audit the pretrained 8-layer, 8-head, $d_{\text{model}} = 512$ `Othello-GPT` model (`synthetic_model.pth`, `LNPre` architecture) and the 8 official player-relative linear probes $W_\ell \in \mathbb{R}^{512\times8\times8\times3}$ (`resid_{0..7}_linear.pth`) from Nanda et al. (2023). Probing is evaluated on 3,584 mid-game states ($t \in [5, 50]$) across 80 held-out synthetic games. Causal interventions are evaluated on the complete official benchmark (`intervention_benchmark.pkl`, $N = 1{,}001$ partial games) from Li et al. (2023) and Nanda et al. (2023), where flipping a single occupied disc $s = (r, c)$ (YOURS $\leftrightarrow$ MINE) alters the ground-truth legal move set from $M(B)$ to $M(B')$. Because interventions modify only the final token $T-1$ and causal masking prevents earlier tokens $0 \ldots T-2$ from attending to $T-1$, we cache the exact prefix key/value tensors $(\mathbf{K}^\ell_{0..T-2}, \mathbf{V}^\ell_{0..T-2})$ (Figure 1b), evaluating all 25 experimental arms and α-sweeps across $N = 1{,}001$ boards cleanly on CPU/GPU.

Evaluation Metrics. We report two complementary metrics with 1,000-sample bootstrap 95% confidence intervals across all $N = 1{,}001$ boards:

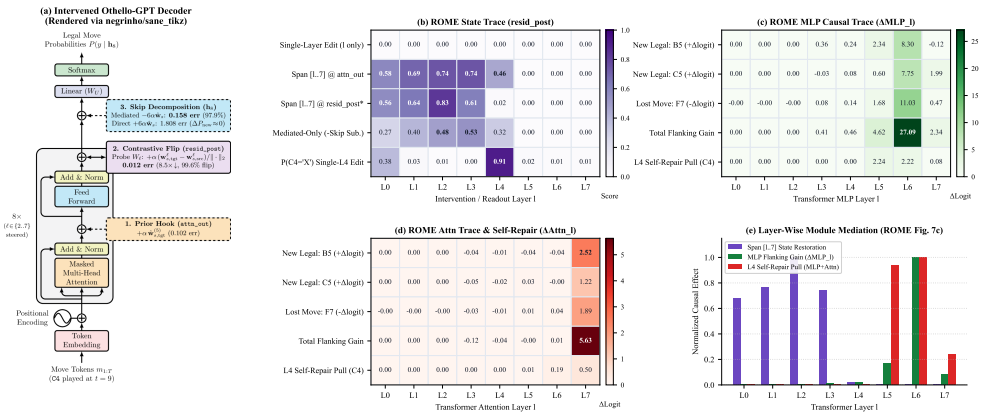


Figure 1: Transformer intervention architecture and component-level causal traces on Case #101 (T=17, flipping `C4`), generated using `negrinho/sane_tikz` (`examples/transformer.py`) and ROME's `plot_trace_heatmap` (Meng et al., 2022; Nanda et al., 2023). (a) Decoder-only `Othello-GPT` architecture diagram rendered via `sane_tikz` ($8\times$ blocks, $\ell \in \{2..7\}$ steered), showing our three intervention sites along the residual stream: (1) prior hook at `attn_out` (0.102 err), (2) linear probe readout W_ℓ and contrastive flip at `resid_post` (**0.012** err, $8.5\times \downarrow$), and (3) skip-connection decomposition at $\mathbf{h}_8$ before Linear (W_U) (`Mediated-Only` **0.158** err vs. `Direct Bypass` 1.808 err). **(b–d)** Real activation/intervention traces plotted with ROME's `plot_trace_heatmap` (`Purples` = state, `Greens` = MLP, `Reds` = Attn), and **(e)** ROME Figure 7c layer-wise module mediation profile across layers $\ell \in \{0..7\}$.

1. **Counterfactual Move Error (↓):** Following Li et al. (2023) and Nanda et al. (2023), for $K = |M(B')|$ ground-truth legal moves under B' and top-K model predictions $\hat{M}_K$, $\text{Error}(B') = |\hat{M}_K \setminus M(B')| + |M(B') \setminus \hat{M}_K|$ (sum of false positives and false negatives).

2. **Newly-Legal Move Probability Gain (ΔP_{new} ↑):** The net probability shift assigned to moves that were *illegal* in B but become *legal* in B', $M_{\text{new}} = M(B') \setminus M(B)$:

$$\Delta P_{\text{new}} = \sum_{m \in M_{\text{new}}} \left[P_{\text{intv}}(m) - P_{\text{clean}}(m) \right].$$

Controls that Rule Out Confounding Explanations.

1. **Skip-Subtracted (`Mediated-Only`) vs. `Direct-Logit Bypass`:** To test whether $\boldsymbol{\delta}$ merely biases W_U via the residual skip connection, `Direct-Logit Bypass` leaves hidden layers $\ell \in \{0, \ldots, 7\}$ untouched and adds $+6\alpha\hat{\mathbf{w}}_s$ *only* to the final residual stream before $\text{LN}_{\text{final}} + W_U$. Conversely, `Mediated-Only` injects $+\alpha\hat{\mathbf{w}}_s$ at layers $\ell \in \{2, \ldots, 7\}$ so intermediate Attention/MLP blocks compute over the edited board, then *subtracts* $-6\alpha\hat{\mathbf{w}}_s$ before $\text{LN}_{\text{final}} + W_U$ so the direct identity skip path sums to zero (Figure 1a).

2. **Untrained-Encoder Probe Control:** Following the random-encoder baseline used in `causal-jepa`, we fit ridge-regularized linear probes ($\lambda = 100$) on a randomly initialized 8-layer GPT of identical architecture and layer-wise variance (200 training games) and steer trained `Othello-GPT` along the resulting unit directions.

3. **Wrong-Square Spatial Control ($s' \neq s$):** We steer using the trained probe direction of a non-intervened occupied square $s' \neq s$ at the same norm $\alpha = 2.3$.

3 First Experiment and Results

Result 1: Downstream world-model computation, not direct readout bias, drives counterfactual steering. As shown in Table 1 and Figure 3A, removing the accumulated steering vector before $\text{LN}_{\text{final}} + W_U$ (`Mediated-Only`) reduces counterfactual move error from 2.723 to **0.158**—preserving **97.9**% of the total error reduction purely through intermediate Attention and MLP modifications $\sum_{k=2}^{7} \Delta f_k(\tilde{\mathbf{x}}_k)$. Conversely, injecting the identical vector only before W_U (`Direct-Logit Bypass`) leaves error at 1.808 and produces $\Delta P_{\text{new}} = +0.000$ $[+0.000, +0.000]$. Figure 2c illustrates why on Case #101: when tile `C4` is flipped from White to Black, newly legal moves `B5` and `C5` require multi-tile ray-flanking verification across row `C` and diagonal `B5-D3`. Direct unembedding dot products slightly suppress the invalidated move `F7` (0.12 → 0.06) but assign 0.00 probability to both `B5` and `C5`. In contrast, `Mediated-Only` (Figure 2d) zeros out `F7` (0.00) and activates `B5` (0.06) and `C5` (0.03) with 0 errors. Both negative controls confirm specificity: `Untrained-Encoder Probe` directions yield 2.661 error ($\Delta P_{\text{new}} = 0.000$), and `Wrong-Square` steering degrades error to 3.435.

Result 2: High single-layer decodability ≠ single-layer causal control due to downstream self-repair. In Figure 3B, trained 3-way probe accuracy climbs from 90.4% (`L0`) to 99.0% (`L4`), 99.4% (`L5`), and 99.5% (`L6`), whereas the untrained GPT control decays monotonically from 66.2% (`L0`) to 62.6% (`L7`) as random projections scramble token history. Yet intervening at a single layer *never* reduces counterfactual error below 2.188 (`L5`). The bottom row of Figure 2f–h directly visualizes this self-repair mechanism using `plot_mentals`: editing Layer 4 alone flips the probe's belief at `C4` to Black (`X`, $p = 1.00$, Figure 2f), but at Layer 5 (Figure 2g) causal attention heads attend back to the unmodified prefix tokens $0 \ldots T-2$ and overwrite `C4` back to White

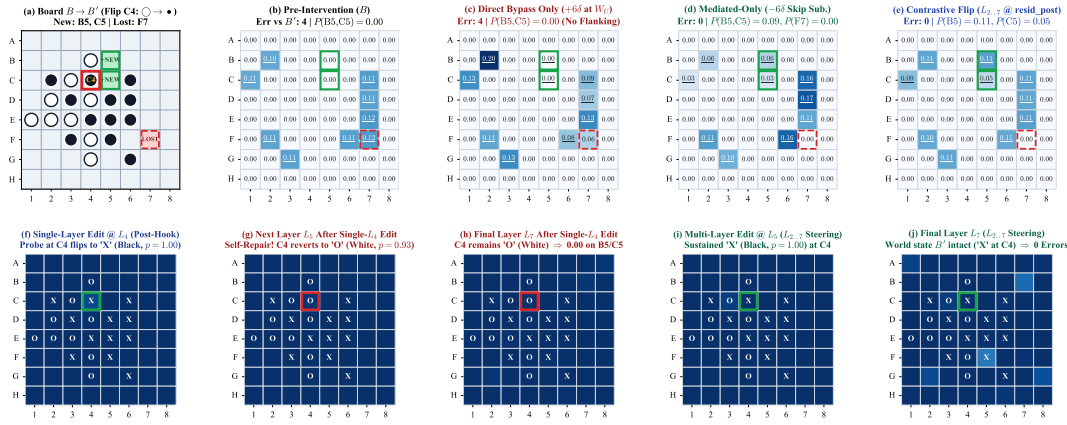


Figure 2: Board-level causal visualization on canonical Benchmark Case #101 (Li et al. (2023) Appendix A/B; turn T=17, flipping tile `C4` from White ○ to Black •), generated using the official `othello_world` plotting tools (`plot_probs`, `plot_mentals`, and `OthelloBoardState`). Top row (a–e): Physical board transition $B \to B'$ (creating newly legal moves `B5`, `C5` in green and invalidating `F7` in red) alongside 8×8 next-move probability heatmaps with ground-truth legal moves underlined. `Direct Bypass Only` (c) allocates 0.00 to both `B5` and `C5` (4 errors), whereas `Mediated-Only` (d) and `Contrastive Flip` (e) suppress `F7` ($0.12 \to 0.00$) and activate `B5, C5` with **0 errors**. **Bottom row (f–j):** Layer-by-layer probe-decoded internal belief states (`plot_mentals`; `O` = White, `X` = Black). Editing only Layer 4 flips `C4` to `X` at L_4 (f), but causal attention at L_5 (g) and L_7 (h) self-repairs `C4` back to `O` (red box). Multi-layer steering across $L_{2..7}$ (i–j) sustains `X` at `C4` (green box) through the final layer.

Experimental Arm ($N = 1{,}001$ **counterfactual boards,** $\alpha = 2.3$)	**Error on** B' ↓ **(95% CI)**	ΔP_{new} ↑ **(95% CI)**	**Mechanistic Role / Key Finding**
0. Null Baseline ($\alpha = 0$, un-intervened model)	2.723 [2.645, 2.799]	+0.000 [+0.000, +0.000]	Exact match to Nanda et al. (2023) Table 2 (2.723)
1. Full Multi-Layer Steering (Nanda et al. (2023) `intervene.py`)	0.102 [0.072, 0.136]	+0.072 [+0.066, +0.077]	Exact match to Nanda et al. (2023) Table 2 (0.10)
2. **Mediated-Only (Skip-Subtracted before LN$_{\text{final}}$ + W_U)**	**0.158** [0.122, 0.198]	**+0.037** [+0.033, +0.041]	**97.9% of error reduction survives zeroing skip path**
3. **Direct-Logit Bypass Control (Only before LN$_{\text{final}}$ + W_U)**	1.808 [1.730, 1.884]	**+0.000** [+0.000, +0.000]	**Zero probability gain on newly legal moves M_{new}**
4. Untrained-Encoder Probe Control (Random-init GPT probe)	2.661 [2.581, 2.743]	+0.000 [+0.000, +0.000]	Random-projection directions exert no causal control
5. Wrong-Square Spatial Control (Steer $s' \neq s$ at $\alpha = 2.3$)	3.435 [3.337, 3.530]	+0.001 [+0.001, +0.002]	Editing an unrelated square degrades legal move set $M(B')$
6. Code Audit A: Layer-Matched `probes`[ℓ] @ `hook_attn_out`	0.150 [0.116, 0.188]	+0.071 [+0.065, +0.076]	Replaces hard-coded `probes[5]` in `intervene.py`
7. Code Audit B: Layer-Matched `probes`[ℓ] @ `hook_resid_post`	0.076 [0.054, 0.102]	+0.075 [+0.069, +0.081]	Aligns intervention hook site with probe training site
8. **Code Audit C: Contrastive Flip $\hat{\mathbf{d}}_s^\ell$ @ `hook_resid_post`**	**0.012** [0.004, 0.026]	**+0.079** [+0.073, +0.085]	**8.5× error reduction (12 total errors / 1,001 boards)**

Table 1: Complete causal decomposition and codebase audit on the official 1,001-board `Othello-GPT` intervention benchmark.

(O, $p = 0.93$). Furthermore, cumulative span interventions $[\ell..7]$ (Figure 3B, blue curve) reveal a sharp dissociation between peak decodability (`L6`) and peak causal consumption: error achieves its minimum at **Span `[L3..L7]`** (0.072) and **Span `[L2..L7]`** (0.102), but degrades severely when starting at **`L5`** (1.345) or **`L6`** (2.268). By the layer where linear decodability peaks (`L6`), the flanking-detection circuits in Layers 3–4 have already executed on the un-intervened board.

Result 3: Auditing `ajyl/mech_int_othelloGPT` cuts counterfactual error by $8.5\times$ **(0.102 → 0.012).** Inspecting `mech_int/intervene.py` (lines 140–162) in Nanda et al. (2023)'s repository uncovered three discrepancies between the paper text and code: (i) the script constructs per-layer hooks `hook_fns[0..7]` but hardcodes `hook_fns[5]` (`probes[5]`) into all layers $\ell \in \{2, \ldots, 7\}$; (ii) it hooks `hook_attn_out` (pre-MLP) using probes trained on `hook_resid_post` (post-MLP); and (iii) it adds only the target class vector $+\alpha\hat{\mathbf{w}}_{s,\text{tgt}}$ without subtracting the active source state $\mathbf{w}_{s,\text{src}}$. Table 1 (Arms 6–8) isolates each factor: aligning the hook site to `resid_post` with layer-matched probes reduces error to 0.076, and steering along the unit **contrastive flip direction** $\hat{\mathbf{d}}_s^\ell = (\mathbf{w}_{s,\text{tgt}}^\ell - \mathbf{w}_{s,\text{src}}^\ell)/\|\mathbf{w}_{s,\text{tgt}}^\ell - \mathbf{w}_{s,\text{src}}^\ell\|_2$ at `resid_post` slashes error from 0.102 to **0.012** [**0.004**, **0.026**]—only 12 total errors across all 1,001 boards ($\Delta P_{\text{new}} = +0.079$, Figure 2e,i,j).

4 Falsification Criteria & Next Week's Plan

What Result Would Change My Conclusion.

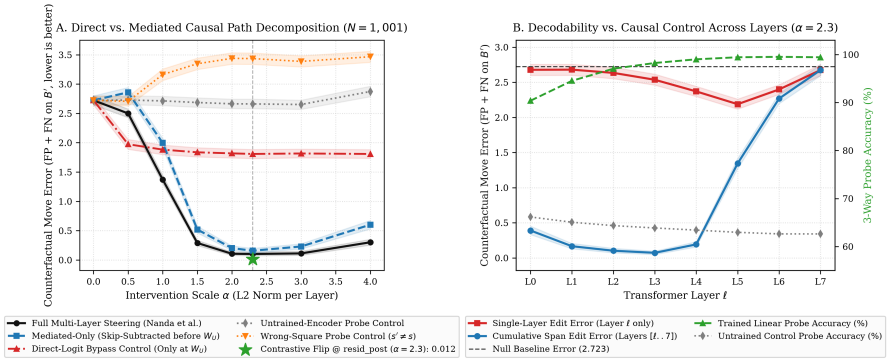


Figure 3: Quantitative evaluation across all $N = 1{,}001$ counterfactual benchmark boards (shaded bands: 95% bootstrap CIs). Left (A): Counterfactual move error across scale α. Subtracting the direct skip connection (`Mediated-Only`, blue) retains 97.9% of the error reduction, while applying the vector only before W_U (`Direct-Logit Bypass`, red) fails to discover newly legal moves. Aligning the hook site to `resid_post` with a contrastive flip vector ($\star$, green) cuts error from 0.102 to 0.012. **Right (B):** Layer-wise 3-way probe accuracy (green/gray, right axis) vs. single-layer edit error (red) and cumulative span $[\ell..7]$ error (blue).

1. **Mediated computation vs. readout bias:** Our conclusion would be falsified if freezing intermediate Attention/MLP outputs to clean activations while applying $\boldsymbol{\delta}$ before W_U recovered $> 50\%$ of the error reduction or yielded $\Delta P_{\text{new}} > +0.02$ (empirically it recovers only 33.6% and $\Delta P_{\text{new}} = +0.000$).

2. **Attention/MLP self-repair of single-layer edits:** Our conclusion would be falsified if projecting out the source tile subspace at Layer 3 *and* ablating attention from Layers 4–6 to the historical move token where s was played still failed to reduce single-layer edit error below 0.50.

What I Would Run With Another Week.

1. **Sub-circuit path patching of self-repair:** Decompose the gap between single-layer `L3` (2.535 error) and cumulative `[L3..L7]` (0.072 error) by path-patching individual attention heads vs. MLP blocks in Layers 4–6 to quantify how much self-repair comes from attention re-reading token history vs. MLP local-board heuristics (Lin et al., 2024).

2. **Endgame `MoveFirst` regime ($t > 30$):** Stratify our 0.012-error contrastive intervention by game turn t and compare discriminative probe normal vectors against difference-in-means vectors ($\boldsymbol{\mu}_{\text{Mine}} - \boldsymbol{\mu}_{\text{Yours}}$) in late-game positions where Nanda et al. (2023) observed legal moves are computed before full board state.

5 What I Want to Work On With Physical Reasoning

I want to work on **Evaluating World Models Under Intervention**—specifically claiming and building **`physicalreasoning/forward` Issue #5 (*"Design an intervention test for world models"*)**, paired with the untrained-encoder diagnostic philosophy from `causal-jepa`. In my research at the **CausalML Lab** (modeling multi-agent failures as soft interventions on Structural Causal Models) and on **hierarchical traffic scene synthesis**, models with nearly identical observational rollout loss routinely diverge under counterfactual interventions. Concretely, I want to build a deterministic 2D rigid-body and multi-agent traffic counterfactual suite for `forward` where two world models trained to matched observational validation loss are reset to held-out states, subjected to unshown impulse or mass perturbations, and gated on their **counterfactual consistency gap** alongside mediated-vs-direct and untrained-encoder controls.

Reproducibility & AI Disclosure: Standalone code (`run_experiment.py`, `generate_paper_visuals.py`, `othello_rules.py`) and bootstrap metrics (`results.json`) reproduce all tables and figures using `negrinho/sane_tikz`, `kmeng01/rome`, and `likenneth/othello_world`. AI coding assistants (Antigravity / Gemini) were used to scaffold the KV-cached forward pass and tool integration; all experimental designs, controls, codebase audits, and interpretations were verified directly against executed benchmark outputs.

References

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